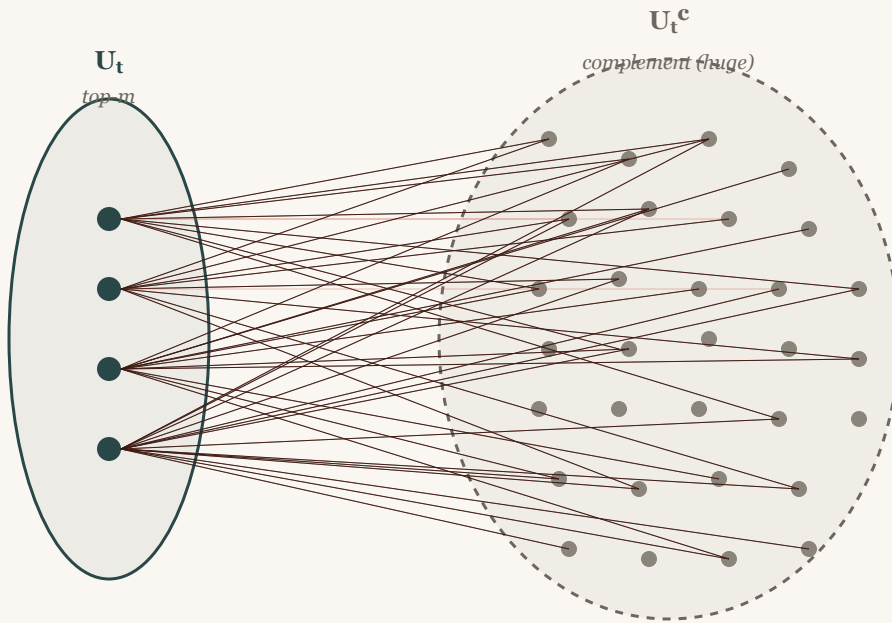


CASE — pruning the comparison space

Why the same statistical guarantees can be achieved with far less work

CLASSICAL • LinGapE / LinGIFA

every top-m arm compared to every other arm

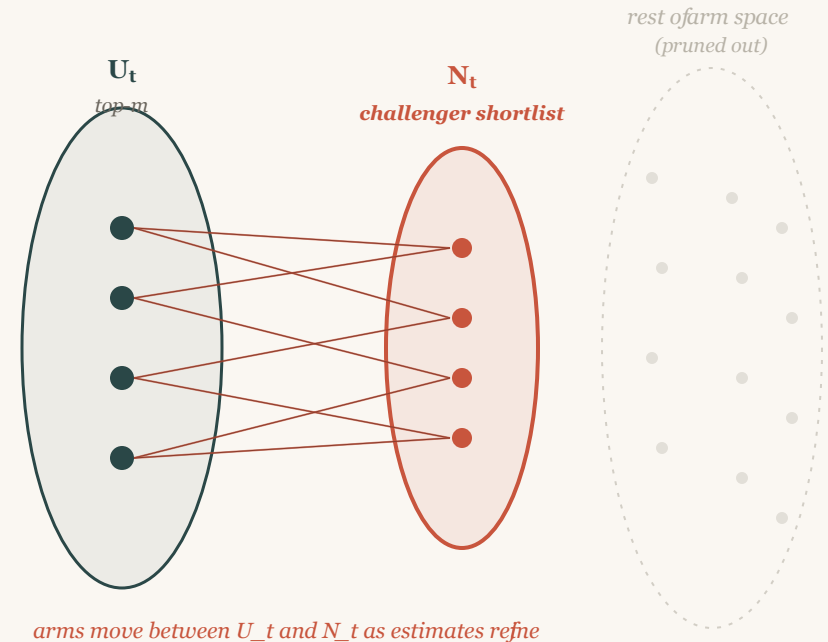


COMPARISON COST

$O(m \times |U_t^c|)$ comparisons per round

CASE • OUR APPROACH

each top-m arm compared only to a challenger shortlist



arms move between U_t and N_t as estimates refine

COMPARISON COST

$O(m \times |N_t|)$ comparisons $\cdot |N_t| \ll |U_t^c|$

The shortlist is maintained adaptively. CASE preserves convergence guarantees of LinGapE while pruning out the comparisons that don't matter.